



Application	Material	Distance	Environment	Preferred Brand	Mode
Operator Safety	Human	5000 mm	Clean	Any	AI Expert Mode

AI Confidence & Reasoning

92%

AI Confidence

For Operator Safety applications requiring detection of humans at a distance of 5000 mm in a clean environment, Safety Light Curtains certified to PLd/SIL2 or higher are the most suitable choice. The current Omron F3SJ, likely a photoelectric sensor, may not provide the necessary safety certification or detection range for this application. A thorough analysis of the operating environment and required protection class led to the selection of Safety Light Curtains from reputable manufacturers.

TOP 3 AI-RECOMMENDED SENSORS

Parameter	Best Match	Alternative 1	Alternative 2
MANUFACTURER	SICK	Banner Engineering	Leuze
MODEL NUMBER	deTec4 Core	SGS	RSL 400
SENSOR TYPE	Safety Light Curtain	Safety Light Curtain	Safety Laser Scanner
DETECTION RANGE	0.5 m to 21 m	0.5 m to 12 m	0.1 m to 8 m
IP RATING	IP67	IP67	IP67
OUTPUT TYPE	OSSD PNP Safety	OSSD PNP Safety	OSSD PNP Safety
EST. PRICE	₹ 85,000	₹ 60,000	₹ 1,20,000
STRENGTHS	High-resolution detection, flexible configuration options, robust design	Easy configuration, compact design, cost-effective	High-precision detection, flexible configuration options, compact design
LIMITATIONS	Complex setup and configuration, higher cost compared to other options	Limited detection range compared to other options, lower resolution	Higher cost compared to other options, complex setup and configuration
DATASHEET	https://www.sick.com/de/en/product-family/detec4-core/c/g917791	https://www.bannerengineering.com/us/en/products/safety-light-curtains.html	https://www.leuze.com/en/usa/products/rsl_400_safety_laser_scaner_844.php

AI Engineering Analysis

RISK ASSESSMENT

The primary risk in this application is the potential for human injury or fatality due to inadequate detection and response. Environmental limitations such as dust, vibration, or extreme temperatures may also impact sensor performance. Regular maintenance and inspection are crucial to ensure sensor reliability and effectiveness.

INSTALLATION NOTES

Mounting and alignment of the Safety Light Curtain are critical to ensure accurate detection and reliable operation. The sensor should be installed in a manner that prevents tampering or accidental damage, and regular checks should be performed to verify proper alignment and function.

MAINTENANCE RECOMMENDATIONS

Regular inspection and maintenance of the Safety Light Curtain are essential to ensure continued reliable operation and compliance with safety standards. This includes cleaning the sensor, checking for damage or wear, and verifying proper function through diagnostic tests.